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Branch :	CMPN/B

Code	<pre> R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ > # --- Lecture 40: Substitution and Replacement of Strings --- > > y = "Number of participants: 25" > sub("25", "30", y) [1] "Number of participants: 30" > y > y = "Mr. Singh is the smart one. Mr. Singh is funny, too." > y [1] "Mr. Singh is the smart one. Mr. Singh is funny, too." > sub("Mr. Singh", "Professor Jha", y) [1] "Professor Jha is the smart one. Mr. Singh is funny, too." > gsub("Mr. Singh", "Professor Jha", y) [1] "Professor Jha is the smart one. Professor Jha is funny, too." > sub("Mr. Singh", "Professor Jha", y) [1] "Professor Jha is the smart one. Mr. Singh is funny, too." > str = c("R Course", "exercises", "include examples of R language") > grep("ex", str, value=TRUE) [1] "exercises" > grep("ex", str, value=FALSE) [1] 2 3 > > grep("R", str, ignore.case=FALSE, value=TRUE) [1] "R Course" > grep("R", str, ignore.case=TRUE, value=TRUE) [1] "R Course" > str = c("R Course", "exercises", "include examples of R language") > grep("R", str, ignore.case=TRUE, value=FALSE) [1] 1 2 3 > grep("R", str, ignore.case=FALSE, value=FALSE) [1] 1 3 > x = "R course 24.07.2021" > y = "Number of participants: 25" > c(x,y) [1] "R course 24.07.2021" > grep("our", c(x,y)) [1] 1 > > grep("Num", c(x,y)) [1] 2 > str = c("R Course", "exercises", "include examples of R language") > grepl("R", str) [1] TRUE FALSE TRUE Error in grepl("ex", str, value = TRUE) : unused argument (value = TRUE) Show Traceback Rerun with Debug > library(MASS) > painters Composition Drawing Colour Expression School Da Uddine 10 8 16 3 A Da Vinci 15 16 4 14 A Del Piombo 8 13 16 7 A Del Sarto 12 16 9 8 A Fr. Penni 0 15 8 0 A Guilio Romano 15 16 4 14 A Michelangelo 8 17 4 8 A Perino del Vaga 15 16 7 6 A Perugino 4 12 10 4 A Raphael 17 18 12 18 A F. Zucarro 10 13 8 8 B Fr. Salviata 13 15 8 8 B Parmigiano 10 15 6 6 B Primaticcio 15 14 7 10 B T. Zucarro 13 14 10 9 B Volterra 12 15 5 8 B Barocci 14 15 6 10 C Cortona 16 14 12 6 C Josepin 10 10 6 2 C L. Jordaens 13 12 9 6 C Testa 11 15 0 6 C Vanius 15 15 12 13 C Bassano 6 8 17 0 D Bellini 4 6 14 0 D Giorgione 8 9 18 4 D Murillo 6 8 15 4 D Palma Giovane 12 9 14 6 D Palma Vecchio 5 6 16 0 D Pordenone 8 14 17 5 D Tintoretto 15 14 16 4 D Titian 12 15 18 6 D Veronese 15 10 16 3 D Albani 14 14 10 6 E Caravaggio 6 6 16 0 E Correggio 13 13 15 12 E Domenichino 15 17 9 17 E Guercino 18 10 10 4 E Lanfranco 14 13 10 5 E R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ > grepl("ex", str, value=TRUE) Console Terminal Background Jobs R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ The Carraci 15 17 13 13 E Durer 8 10 10 8 F Holbein 9 10 16 13 F Pourbus 4 15 6 6 F Van Leyden 8 6 6 4 F Diepenbeck 11 10 14 6 G J. Jordaens 10 8 16 6 G Otho Venius 13 14 10 10 G Rembrandt 15 6 17 12 G Rubens 18 13 17 17 G Teniers 15 12 13 6 G Van Dyck 15 10 17 13 G Bourdon 10 8 8 4 H Le Brun 16 16 8 16 H Le Suer 15 15 4 15 H Poussin 15 17 6 15 H > rownames(painters) [1] "Da Uddine" "Del Piombo" "Del Sarto" [5] "Fr. Penni" "Guilio Romano" "Michelangelo" [9] "Perugino" "Raphael" "F. Zucarro" "Fr. Salviata" [13] "Parmigiano" "Primaticcio" "T. Zucarro" "Volterra" [17] "Barocci" "Cortona" "Josepin" "L. Jordaens" [21] "Testa" "Vanius" "Bassano" "Bellini" [25] "Giorgione" "Murillo" "Palma Giovane" "Palma Vecchio" [29] "Pordenone" "Tintoretto" "Titian" "Veronese" [33] "Albani" "Caravaggio" "Correggio" "Domenichino" [37] "Guercino" "Lanfranco" "The Carraci" "Durer" [41] "Holbein" "Pourbus" "Van Leyden" "Diepenbeck" [45] "J. Jordaens" "Otho Venius" "Rembrandt" "Rubens" [49] "Teniers" "Van Dyck" "Bourdon" "Le Brun" [53] "Le Suer" "Poussin" > > is.numeric(painters\$School) [1] FALSE > is.numeric(painters\$Drawing) [1] TRUE > > is.factor(painters\$School) [1] TRUE > is.factor(painters\$Drawing) [1] FALSE > > colnames(painters) [1] "Composition" "Drawing" "Colour" "Expression" "School" </pre>
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ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

> summary painters

CompositionDrawingColourExpressionSchool

Min. : 0.00 Min. : 6.00 Min. : 0.00 Min. : 0.000 A :10

1st Qu.: 8.25 1st Qu.:10.00 1st Qu.: 7.25 1st Qu.: 4.000 D :10

Median :12.50 Median :13.50 Median :10.00 Median : 6.000 E : 7

Mean :11.56 Mean :12.46 Mean :10.94 Mean : 7.667 G : 7

3rd Qu.:15.00 3rd Qu.:15.00 3rd Qu.:16.00 3rd Qu.:11.500 B : 6

Max. :18.00 Max. :18.00 Max. :18.00 Max. :18.000 C : 6

(Other): 8

> library(MASS)

> painters

CompositionDrawingColourExpressionSchool

Da Uline108163A

Da Vinci1516414A

Del Plombo813167A

Del Sarto121698A

Fr. Penni01580A

Guilio Romano1516414A

Michelangelo81748A

Pertino del Vaga151676A

Perugino412104A

Raphael17181218A

F. Zucarro101388B

Fr. Salviata131588B

Parmigiano101566B

Primaticcio1514710B

T. Zucarro1314109B

Volterra121558B

Barocci1415610C

Cortona1614126C

Josepin101062C

L. Jordaens131296C

Testa111506C

Vanius15151213C

Bassano68170D

Bellini46140D

Giorgione89184D

Murillo68154D

Palma Giovane129146D

Palma Vecchio56160D

Pordenone814175D

Tintoretto1514164D

Titian1215186D

Veronese1510163D

Albani1414106E

Caravaggio66160E

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

Correggio13131512E

Domenichino1517917E

Guercino1810104E

Lanfranco1413105E

The Carracci15171313E

Durer810108F

Holbein9101613F

Pourbus41566F

Van Leyden8664F

Diepenbeck1110146G

J. Jordaens108166G

Otho Venius13141010G

Rembrandt1561712G

Rubens18131717G

Teniers1512136G

Van Dyck15101713G

Bourdon10884H

Le Brun1616816H

Le Suer1515415H

Poussin1517615H

> summary painters\$School

A B C D E F G H

10 6 6 10 7 4 7 4

>

> attach(painters)

> summary(School)

A B C D E F G H

10 6 6 10 7 4 7 4

> summary(Composition)

Min. 1st Qu. Median Mean 3rd Qu. Max.

0.00 8.25 12.50 11.56 15.00 18.00

> detach(painters)

>

> # After detach(), use painters\$School again.

> summary(painters\$School)

A B C D E F G H

10 6 6 10 7 4 7 4

> subset(painters, School=="F")

CompositionDrawingColourExpressionSchool

Durer810108F

Holbein9101613F

Pourbus41566F

Van Leyden8664F

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/
> painters[painters[["School"]] == "F", ]
      Composition Drawing Colour Expression School
Durer             8      10      10           8      F
Holbein           9      10      16          13      F
Pourbus           4      15       6           6      F
Van Leyden        8       6       6           4      F
> subset(painters, Composition <= 6)
      Composition Drawing Colour Expression School
Fr. Penni         0      15       8           0      A
Perugino          4      12      10           4      A
Bassano           6       8      17           0      D
Bellini           4       6      14           0      D
Murillo           6       8      15           4      D
Palma Vecchio     5       6      16           0      D
Caravaggio        6       6      16           0      E
Pourbus           4      15       6           6      F
> subset(painters, School=="F", select=c(-3,-5))
      Composition Drawing Expression
Durer             8      10           8
Holbein           9      10          13
Pourbus           4      15           6
Van Leyden        8       6           4
>
> splitted = split(painters, painters$School)

```

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/
> splitted
$A
      Composition Drawing Colour Expression School
Da Udine          10       8      16           3      A
Da Vinci          15      16       4          14      A
Del Piombo         8      13      16           7      A
Del Sarto          12      16       9           8      A
Fr. Penni          0      15       8           0      A
Gulio Romano       15      16       4          14      A
Michelangelo        8      17       4           8      A
Perino del Vaga     15      16       7           6      A
Perugino           4      12      10           4      A
Raphael           17      18      12          18      A

$B
      Composition Drawing Colour Expression School
F. Zucarro         10      13       8           8      B
Fr. Salviata       13      15       8           8      B
Parmigiano         10      15       6           6      B
Primaticcio        15      14       7          10      B
T. Zucarro         13      14      10           9      B
Volterra           12      15       5           8      B

$C
      Composition Drawing Colour Expression School
Barocci            14      15       6          10      C
Cortona            16      14      12           6      C
Josepin            10      10       6           2      C
L. Jordaens        13      12       9           6      C
Testa              11      15       0           6      C
Vanius             15      15      12          13      C

$D
      Composition Drawing Colour Expression School
Bassano            6       8      17           0      D
Bellini            4       6      14           0      D
Giorgione          8       9      18           4      D
Murillo            6       8      15           4      D
Palma Giovane      12       9      14           6      D
Palma Vecchio      5       6      16           0      D
Pordenone          8      14      17           5      D
Tintoretto         15      14      16           4      D
Titian             12      15      18           6      D
Veronese           15      10      16           3      D

```

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/

$E
      Composition Drawing Colour Expression School
Albani          14      14      10          6      E
Caravaggio       6       6      16          0      E
Correggio        13      13      15         12      E
Domenichino      15      17       9         17      E
Guerchino        18      10      10          4      E
Lanfranco        14      13      10          5      E
The Carracci     15      17      13         13      E

$F
      Composition Drawing Colour Expression School
Durer           8       10      10          8      F
Holbein          9       10      16         13      F
Pourbus          4       15       6          6      F
Van Leyden       8       6       6          4      F

$G
      Composition Drawing Colour Expression School
Diepenbeck       11      10      14          6      G
J. Jordaens      10       8      16          6      G
Otho Venius      13      14      10         10      G
Rembrandt       15       6      17         12      G
Rubens           18      13      17         17      G
Teniers          15      12      13          6      G
Van Dyck         15      10      17         13      G

$H
      Composition Drawing Colour Expression School
Bourdon          10       8       8          4      H
Le Brun          16      16       8         16      H
Le Suer          15      15       4         15      H
Poussin          15      17       6         15      H

```

```
> is.data.frame(splittedSA)
```

```
[1] TRUE
```

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/

> df1=data.frame(state=c("UP", "MP", "AP", "JK"),
+               popsize=c(1000,2000,3000,4000))
>
> df2=data.frame(state=c("UP", "MP", "AP", "JK"),
+               samplesize=c(100,200,300,400),
+               surveycompleted=c("Yes", "No", "Yes", "No"))
>
> df1
  state popsize
1  UP    1000
2  MP    2000
3  AP    3000
4  JK    4000
> df2
  state samplesize surveycompleted
1  UP         100           Yes
2  MP          200            No
3  AP          300           Yes
4  JK          400            No
> cbind(df1,df2)
  state popsize state samplesize surveycompleted
1  UP    1000   UP         100           Yes
2  MP    2000   MP          200            No
3  AP    3000   AP          300           Yes
4  JK    4000   JK          400            No
> merge(df1,df2,by="state")
  state popsize samplesize surveycompleted
1  AP    3000         300           Yes
2  JK    4000         400            No
3  MP    2000         200            No
4  UP    1000         100           Yes
>
> # rbind() - stack data frames vertically
> df11=data.frame(state=c("UP", "MP", "AP", "JK"),
+               popsize=c(1000,2000,3000,4000))
>
> df22=data.frame(state=c("Bihar", "Delhi", "Punjab"),
+               popsize=c(100,200,300))
>
> df11
  state popsize
1  UP    1000
2  MP    2000
3  AP    3000
4  JK    4000

```

```
> df22
state popsize
1 Bihar 100
2 Delhi 200
3 Punjab 300
> rbind(df11,df22)
state popsize
1 UP 1000
2 MP 2000
3 AP 3000
4 JK 4000
5 Bihar 100
6 Delhi 200
7 Punjab 300
>
```

Environment			History	Connections	Git	Tutorial
R - Global Environment			28 MiB			
Data						
ab	List of 6					
datemat	chr [1:4, 1:3]	"2020" "2021" "2022" "2023" "07" "08" "09" "10" "2..."				
datematrix	num [1:4, 1:3]	2020 2021 2022 2023 7 ...				
datesplt	List of 4					
df1	4 obs. of 2 variables					
df11	4 obs. of 2 variables					
df2	4 obs. of 3 variables					
df22	3 obs. of 2 variables					
list1	List of 6					
list12	List of 6					
list2	List of 6					
splitted	List of 8					
Values						
brands	chr [1:7]	"A" "A" "B" "B" "B" "B" "C"				
brands_fac	Factor w/ 3 levels	"A","B","C": 1 1 2 2 2 2 3				
colours	chr [1:3]	"blue" "green" "red"				
d	"Thu Oct 1 14:08:05 2026"					
dates	chr [1:4]	"2020-07-24" "2021-08-25" "2022-09-26" "2023-10-27"				
evenno	num [1:5]	2 4 6 8 10				
facy	Factor w/ 6 levels	"one","two","three",...: 1 4 3 5 4 2 4				
income	Ord.factor w/ 3 levels	"low"<"medium"<...: 3 3 1 2 2				
labels.dieface	chr [1:6]	"one" "two" "three" "four" "five" "six"				
names	chr [1:3]	"Prof. Singh" "Mr. Venkat" "Dr. Jha"				
possible.dieface	num [1:6]	1 2 3 4 5 6				
str	chr [1:3]	"R Course" "exercises" "include examples of R language"				
x	"R course 24.07.2021"					
y	"Number of participants: 25"					
z	"INDIAN INSTITUTE OF TECHNOLOGY"					
z1	num [1:3]	1.1 2.22 3.33				